

LEXICAL EXCEPTIONALITY IN PARADIGM-SPECIFIC LEARNING: MODELING STEM-FINAL OBSTRUENT ALTERNATIONS IN KOREAN VERBS AND ADJECTIVES

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Question

How do learners identify **exceptional** forms and their relevant paradigms when **multiple repair strategies** [1] coexist to solve the same markedness problem?

Korean Stem-Final Conjugation

When /p/-, /t/-, and /s/-final stems conjugate before vowel-initial suffixes, only the irregular (alternating) class undergoes **intervocalic lenition**. Each paradigm adopts a distinct repair strategy:

- p-stems: /p/ → [w] / V_V
- t-stems: /t/ → [ɾ] / V_V
- s-stems: /s/ → ∅ / V_V

*The alternation-triggering environment is absent before consonant-initial suffixes.

Class	Stem	V_V	V_C
p-alt.	/ka.kap/ 'close'	[ka.ka.w-Λ]	[ka.kap.-ta]
p-non-alt.	/ip/ 'wear'	[i.p-Λ]	[ip.-ta]
t-alt.	/kje.dat/ 'realize'	[kje.da.r-a.jo]	[kje.dat.-ta]
t-non-alt.	/mut/ 'bury'	[mu.t-Λ]	[mut.-ta]
s-alt.	/nas/ 'better'	[na.-a.do]	[nas.-ta]
s-non-alt.	/us/ 'laugh'	[u.s-Λ]	[us.-ta]

The Indexation Approach

Alternating and non-alternating classes require competing generalizations:

- alt. stems: *V-OBS-V $\gg$ FAITH
- non-alt. stems: FAITH $\gg$ *V-OBS-V

[2] resolves this ranking conflict through constraint cloning [3]:

*V-OBS-V_{Alter} $\gg$ FAITH $\gg$ *V-OBS-V_{Non-Alter}

Expanded ranking: Paradigm-specific repairs require different FAITH constraints to be outranked by those associated with other alternation patterns.

*V-OBS-V_P^{Alt} $\gg$ ID-LAT, MAX $\gg$ ID-C $\gg$ *V-OBS-V_P^{NonAlt}

*V-OBS-V_T^{Alt} $\gg$ ID-C, MAX $\gg$ ID-LAT $\gg$ *V-OBS-V_T^{NonAlt}

*V-OBS-V_S^{Alt} $\gg$ ID-C, ID-LAT $\gg$ MAX $\gg$ *V-OBS-V_S^{NonAlt}

Scaled MaxEnt

A probabilistic implementation of constraint indexation [4, 5].

$$\mathcal{H}_i = \sum_{\gamma \in \Gamma} \left(w_{\gamma} + \sum_{m \in \mu_i} s_{\gamma m} \right) v_{\gamma i}$$

(Γ : constraint set; $\mathcal{H}_i$: candidate harmony; w_{γ} : general constraint weight; $s_{\gamma m}$: morpheme-specific scaling factor; $v_{\gamma i}$: violations of γ by candidate i)

The additive penalty $s_{\gamma m}$ plays a role analogous to lexical indexation (*Predictions*):

- (1) *V-OBS-V_{Alter} $\gg$ FAITH $\Rightarrow$ increased scaling for alternating classes
- (2) FAITH constraints implicated in the repair receive the smallest scaling factors:
 - ID-C (/p/-paradigm)
 - ID-LAT (/t/-paradigm)
 - MAX (/s/-paradigm)
- (3) Only lexically idiosyncratic stems receive nonzero scaling; suffixes remain unscaled.

Idealized learning outcome

	*V-OBS-V	MAX	ID-LAT	ID-C
General weights	4	6	6	6
p-alternating	6	3	2	0
p-non-alternating	0	1	1	1
t-alternating	6	3	0	4
t-non-alternating	0	1	1	1
s-alternating	6	0	3	3
s-non-alternating	0	1	1	1
suffixes	0	0	0	0

Experiment

Toy stems were generated in CV-/p,t,s/ forms, reflecting Korean syllable phonotactics. Frequencies were modeled after the distribution reported in [2].

Suffix inventories were constructed by progressively adding paired -V and -CV suffixes, each either triggering or failing to trigger lenition.

Stem statistics	Toy suffix inventories
p-alt 68	Condition Suffix inventory
p-non-alt 5	2-suffix -ta, -a
t-alt 5	4-suffix -ta, -a, -te, -e
t-non-alt 7	6-suffix -ta, -a, -te, -e, -tu, -u
s-alt 5	8-suffix -ta, -a, -te, -e, -tu, -u, -to, -o
s-non-alt 4	10-suffix -ta, -a, -te, -e, -tu, -u, -to, -o, -ti, -i

Results

Scaling weights increasingly converge on the locus of exceptionality as the number of suffix types increases:

Stage 1: Scaling weights are misattributed to non-idiosyncratic suffixes.

2-suffix condition

	*V-OBS-V	MAX	ID-LAT	ID-C
General weights	0.69	2.6	2.6	0
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	0
t-alternating stems	0	0	0	0
t-non-alternating stems	0	0	0	0
s-alternating stems	0	0	0	0
s-non-alternating stems	0	0	0	0
-V suffix	0.69	0	0	0
-CV suffix	0	1.92	1.92	4.51

Stage 2: Scaling weights begin to distinguish alternating and non-alternating stems.

Note: /p/-alternating stems remain unscaled (i.e., fail to be lexicalized).

6-suffix condition

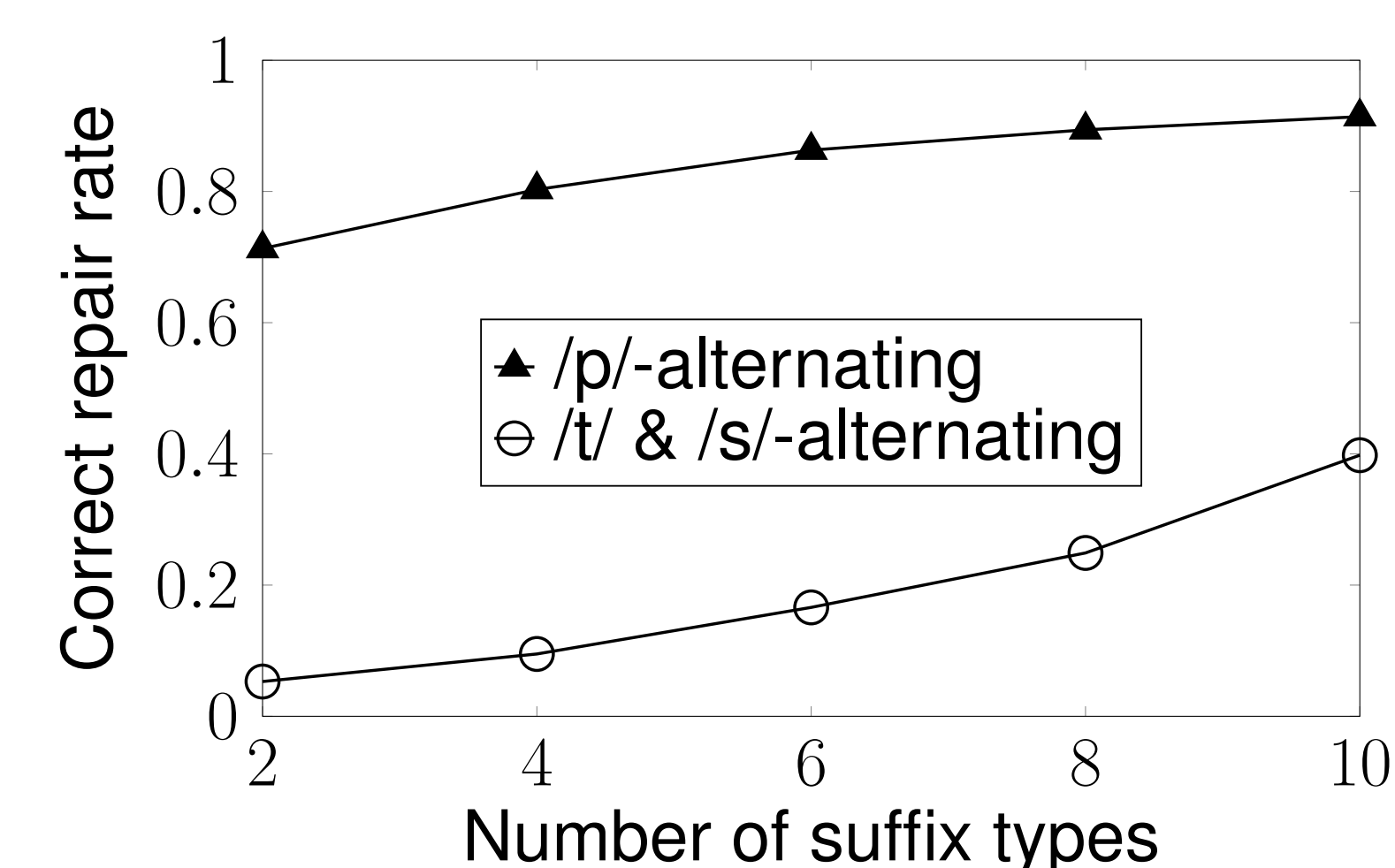
	*V-OBS-V	MAX	ID-LAT	ID-C
General weights	4.98	6.29	6.29	2.71
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	2.54
t-alternating stems	0.61	0	0	2.88
t-non-alternating stems	0	0	0	2.54
s-alternating stems	0.61	0	0	2.88
s-non-alternating stems	0	0	0	2.54
-V suffix	0	0	0	0
-CV suffix	0	0	0	1.53

Stage 3: Paradigm-specific patterns are distinguished among alternating stems.

10-suffix condition

	*V-OBS-V	MAX	ID-LAT	ID-C
General weights	7.11	8.98	8.98	4.46
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	3.76
t-alternating stems	2.55	0.69	0	5.2
t-non-alternating stems	0	0	0	3.76
s-alternating stems	2.55	0	0.69	5.2
s-non-alternating stems	0	0	0	3.76
-V suffix	0	0	0	0
-CV suffix	0	0	0	0

This progression coincides with improved prediction accuracy.



Repair accuracy in intervocalic contexts

- Predictions (1) and (2) emerge gradually in the 4-suffix and 6-suffix conditions, respectively.
- Prediction (3) is satisfied only once sufficient morphological evidence becomes available (10-suffix condition).
- Across all conditions, the /p/ → [w] alternation is absorbed into the general grammar rather than being treated as a lexically idiosyncratic pattern.
 - the /p/-alternating class remains unscaled;
 - ID-C receives the largest scaling weights in the remaining classes.
- Distributional asymmetry between /p/ and /t,s/ alternations is reflected in prediction accuracy, with the former learned more reliably.

Discussion

- The model adopts a *least-cost* solution by treating statistically dominant morpheme classes as the general pattern.
- Sensitivity to lexical statistics: without sufficient morphological evidence, the locus of exceptionality cannot be reliably learned.
- Our empirical simulation suggests that, in naturalistic settings, statistical sensitivity arises from limited paradigm saturation [6] rather than absolute type frequency.

References

- [1] Joe Pater. Austronesian nasal substitution and other nc effects. *The prosody-morphology interface*, pages 310–343, 1999.
- [2] Ponghyung Lee. Constraint cloning and lexical listing in Korean irregular verbs and adjectives. *Studies in Linguistics*, 2013.
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- [4] Tal Linzen, Sofya Kasyanenko, and Maria Gouskova. Lexical and phonological variation in Russian prepositions. *Phonology*, 30(3):453–515, 2013.
- [5] Coral Hugtho, Andrew Lamont, Brandon Prickett, and Gaja Jarosz. Learning exceptionality and variation with lexically scaled maxent. In *Proceedings of the Society for Computation in Linguistics (SCIL)* 2019, pages 91–101, 2019.
- [6] Jordan Kodner. Computational models of morphological learning. In *Oxford Research Encyclopedia of Linguistics*, 2022.

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Code:
https://github.com/stxllastar/scaled_conjugation
Paper:

